Critical Minerals, Geopolitics, and the Green Transition

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Critical minerals will power the green transition

- Lithium, nickel, cobalt, and other minerals
  - Critical for producing advanced batteries
  - Concentrated geographically, but traded globally
- Concentrated endowments give countries market power

Bloomberg

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Markets | Hyperdrive

Indonesia Weighs Deep Cuts to Nickel Mining to Boost Prices

- Energy ministry considering big reduction, people familiar say
- Prices of the battery metal have slumped on booming supply

By Eddie Spence and Faris Mokhtar
December 19, 2024 at 6:55 AM GMT-3

BENCHMARK SOURCE
Supply Chain Intelligence for the Energy Transition

Australian lithium supply cut by 15% due to low spodumene prices

14th November 2024

Fastmarkets

Cobalt export quotas: DRC sets limits to rebalance global supply

The Democratic Republic of Congo has introduced cobalt export quotas following the suspension period, setting limits on shipments while outlining future allocations. The policy is designed to balance global supply and demand while supporting the country's ambition to develop domestic processing capacity.

September 22, 2025

MINING.COM

South America looks at creating "lithium OPEC"

Cecilia Jamasmie | March 6, 2023 | 6:53 am Intelligence News Suppliers & Equipment Latin America Lithium

This paper

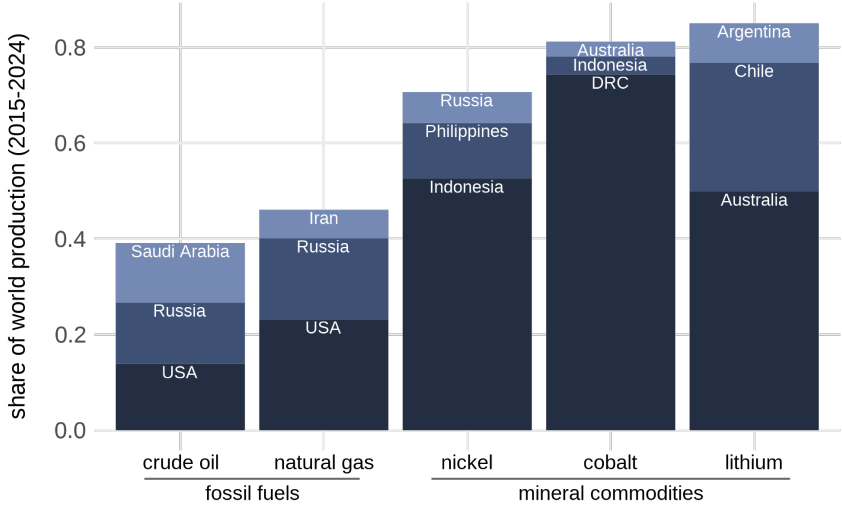
- What are the effects of energy resource concentration?
  - Market power and interdependence (geopolitics)
  - Global battery adoption (green transition)
- Minerals are joint inputs in battery production
  - We need to study these markets jointly
- Dynamic equilibrium model of global mineral markets

Mechanisms

- Market power with complements $\neq$ market power with substitutes
- Complementarity affects interdependence
  - Saudi oil cut $\rightarrow$ US oil gains (substitutes)
  - Indonesian Ni cut $\rightarrow$ Australian Li loses (complements)
- Complementarity affects adoption
  - Within-mineral consolidation $\rightarrow$ raises markups (substitutes)
  - Across-mineral consolidation $\rightarrow$ eliminates double markups (complements)

Background

Mineral supply is geographically concentrated



Critical minerals

- Minerals with low substitutability + high supply risk
  - Focus on criticality for energy transition (lithium, cobalt, nickel)
  - Separate national security purposes (rare earths)

Batteries	%
EV	80
BESS	15
Portable	5

EV batteries	%
Lithium	65
Cobalt	45
Nickel	11

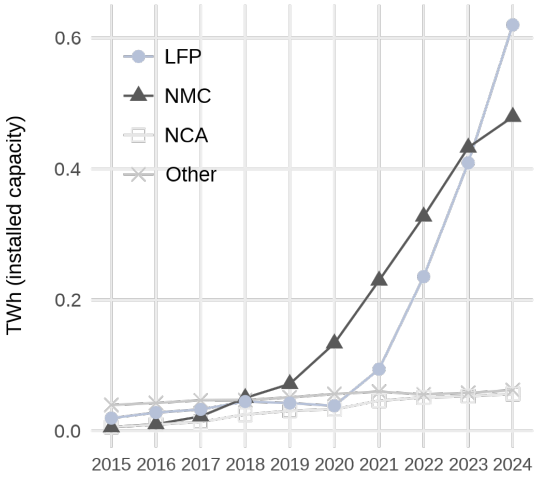
Stages

1. Upstream mining and processing
   - Australia, Indonesia, DRC with Chinese ownership
2. Midstream refining and battery manufacturing
   - China
3. Downstream EV manufacturing
   - China, US, Europe, Korea, Japan

Data

- Mineral supply: mine-level production and costs (GlobalData, S&P, BMI)
  - Near universe of world production for lithium, nickel, cobalt
- Battery demand: EV producer-level consumption (Rho Motion)
  - E.g., Tesla's purchases of NMC batteries for Model S production in US, Q1 2021
- Battery-mineral "recipes" (Argonne National Laboratory)

Mineral recipes vary by battery technology

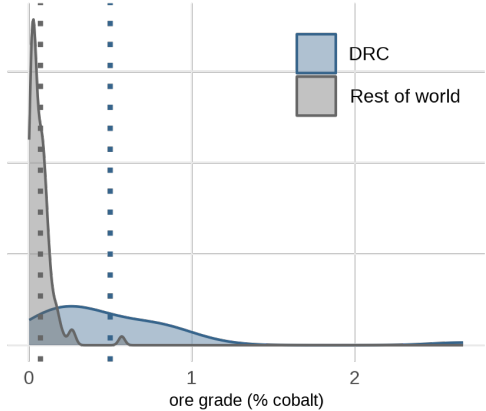


	nickel	lithium	cobalt
LFP	-	0.09	-
NMC	0.47	0.11	0.18
NCA	0.65	0.1	0.12

kg mineral/kWh capacity

With heterogeneous costs across mines





Framework

Market concentration with complementarity

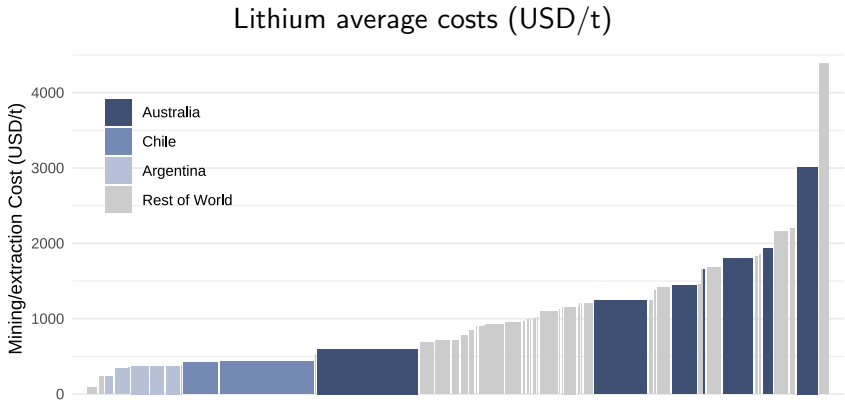
- Indonesia exercises market power for nickel
  - Nickel price ↑, Indonesian profits ↑
  - But consumers switch away from nickel-heavy batteries
- Policy spillovers for Australian lithium
  - Helped via **substitution** toward lithium-heavy batteries
  - Hurt via **complementarity** as nickel-heavy batteries still use lithium
- Green adoption in aggregate
  - Slower if **gross substitutes** (both prices ↑)
  - Ambiguous if **gross complements** (Ni price ↑, Li price ↓)

Empirics

1. Mineral **supply** s^m with dynamic extraction (geopolitics)
2. Battery **demand** d_j across technologies (green transition)
3. **Recipes** r_j^m connect minerals and batteries

Supply

We construct supply curves from our cost data



Supply by mine i, year t

Euler condition	Marginal revenue	Marginal cost
$mr^{it} - mc^{it} = \beta \mathbb{E}^{it}[mr^{it+1} - mc^{it+1}]$	$mr^{it} = p^t g^i$	$mc^{it} = \kappa x^{it} + \varepsilon^{it}$

- **Dynamics** from finite reserves (Hotelling)
  - Extraction x^{it}, price p^t, grade g^i (output $g^i x^{it}$)
- **Cost convexity** κ, which we estimate
  - Cost shocks ε^{it}

Estimation

$$\underbrace{p^t g^i}_{mr^{it}} = \underbrace{\kappa x^{it} + \varepsilon^{it}}_{mc^{it}} + \eta^{it}$$

- To simplify, consider static FOC for estimation
  - Endogeneity problem from cost shocks ε^{it}
  - Need an instrument, which gives an additional moment

Estimation with cost data

$$\underbrace{p^t g^i}_{mr^{it}} = \underbrace{\kappa x^{it} + \varepsilon^{it}}_{mc^{it}} + \eta^{it}, \quad ac^{it} = \frac{\kappa}{2} x^{it} + \varepsilon^{it}$$

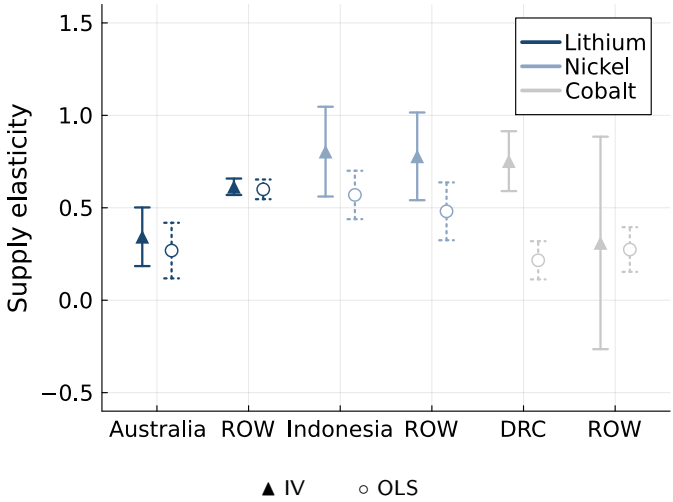
- We use an average cost moment
  - Marginal cost is $\kappa x + \varepsilon$, so total cost is $(\frac{\kappa}{2} x + \varepsilon) x$
  - Invert average cost to isolate ε

Empirical specification

$$\Delta p^{mt} g^i - \Delta ac^{it} = \frac{\kappa^m}{2} \frac{\Delta x^{it}}{\bar{x}^i} + \Delta \tau^{km} g^i + \Delta \tau^{om} g^i + \eta^{it}$$

- With dynamics, use lagged IV for expectational error (Hall 1978)
- Also accounting for capacity $\bar{x}^i$ and policy wedges τ^m

Mineral elasticities



- Australian lithium is less elastic than other lithium producers
- Nickel is more elastic in general
- DRC cobalt is more elastic than other cobalt producers

Demand

Demand by battery j, region k, quarter t

$$w_{jkt} = \alpha_{jt} + \beta_j \log \frac{X_{kt}}{P_t} + \sum_{j'} \gamma_{jj'} \log p_{j't} + \varepsilon_{jkt}$$

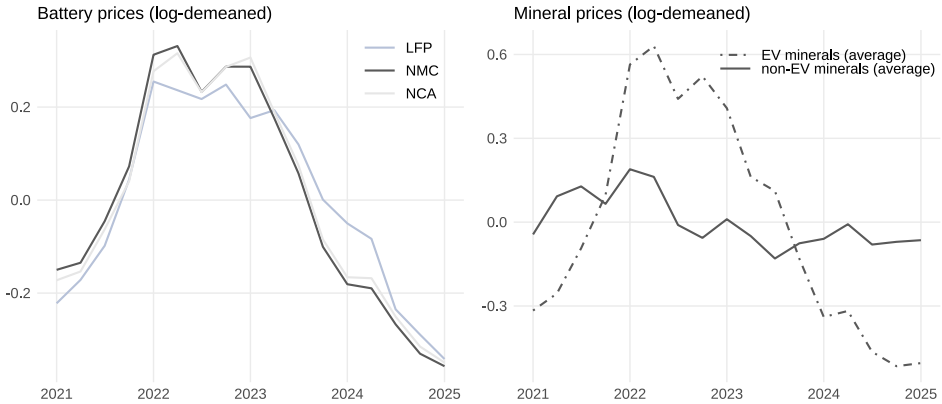
- Almost ideal demand system (Deaton & Muellbauer 1980)
  - Expenditure shares w_{jkt}, prices p_{jt}, translog price index P_t
- Avoids specifying the characteristics that govern substitution
- Endogeneity from demand shocks ε_{jkt}

Estimation with instruments

$$z_{jt} = \sum_{m \in \mathcal{M}'} r_j^m p^{mt}, \quad \mathcal{M}' = \{\text{Al, Mn, Fe, Ph}\}$$

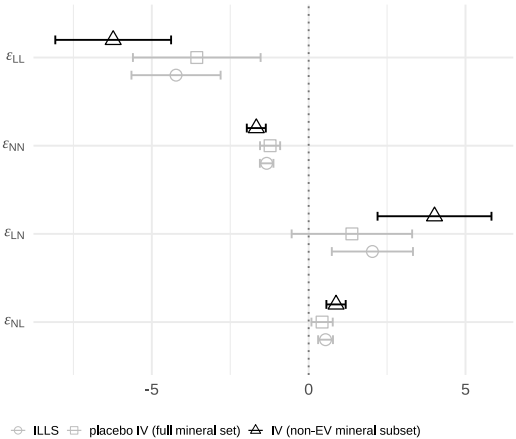
- Secondary mineral prices as cost shifters for battery prices
  - Relevance: minerals affect batteries through recipes r_j^m
  - Exclusion: EVs are less than 3% of demand for $\mathcal{M}'$

Price variation



- Recipe variation disentangles battery prices, which move together
  - Excluded: lithium, nickel, and cobalt, which move with batteries

Battery elasticities



- IV $>$ OLS as expected
  - But only with $\mathcal{M}' \subset \mathcal{M}$
- $LL > NN$ reflects quality of L
  - Shorter driving range
  - More price-sensitive customers
- LN and NL capture switching

Counterfactuals

Equilibrium prices

- Mineral prices clear markets (with non-battery demand δ^{mt})

$$\sum_{i \in \mathcal{I}^{mt}} s^{it} = \sum_{j} r_j^m d_{jt} + \delta^{mt}$$

- Battery prices from mineral prices (with midstream wedge μ_{jt})

$$p_{jt} = \sum_{m} r_j^m p^{mt} + \mu_{jt}$$

- **Equilibrium**: sequence of prices $\{p^{mt}\}$ that clear markets $\forall\, m, t$

Solving the model

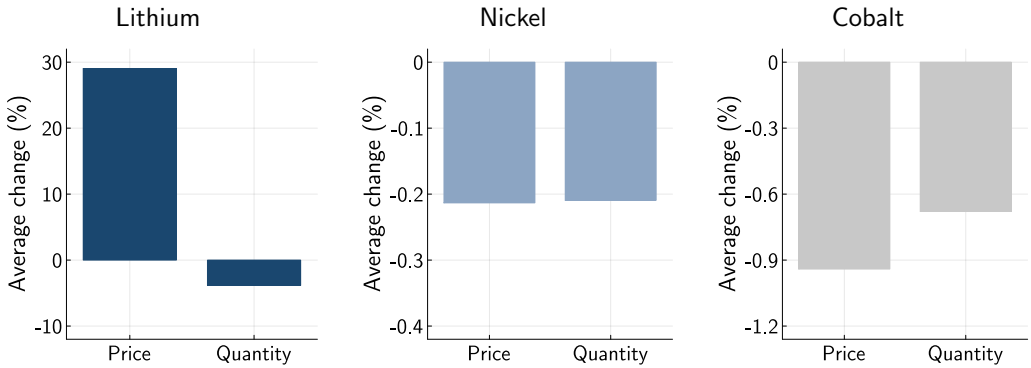
- A country restricts supply to exercise market power
  - Production tax maximizes producer surplus and government revenue
- Solve for equilibrium prices and production over time
  - Other producers respond to the new price path
  - Consumers substitute across battery technologies
- Compare to baseline equilibrium with no restrictive policies

1. Supply chain vulnerability

	Price effects (%)				
	Lithium	Nickel	Cobalt	Battery L	Battery N
Lithium (Australia)	107	−1.09	−3.69	9.47	9.46
Nickel (Indonesia)	−1.84	76.0	−3.43	−0.17	3.55
Cobalt (DRC)	−6.19	−3.74	865	−0.58	14.1

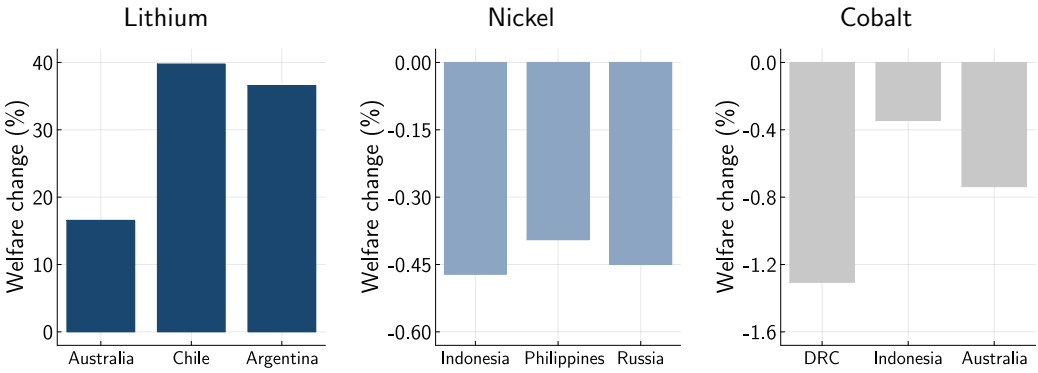
- Remove top producer, solve for equilibrium price changes
  - Large price increases (76% to 865%)
  - Lithium is common input (raises both battery prices)
  - Minerals are **gross complements** (negative cross-effects)

2. Unilateral policy: Australian lithium



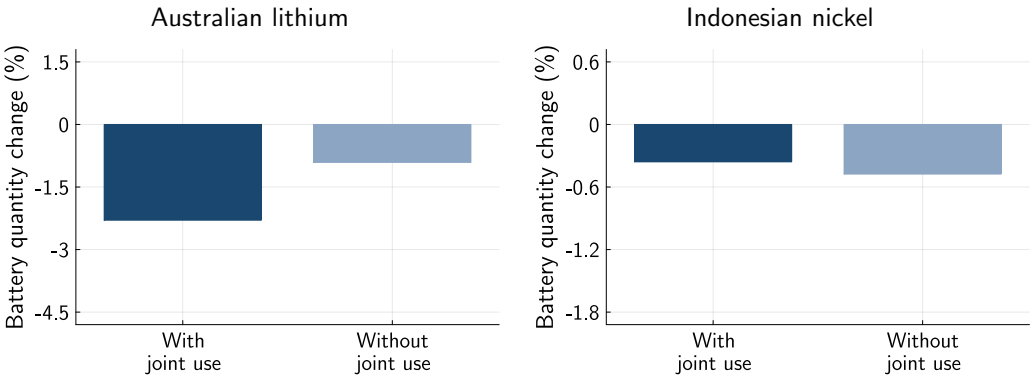
- Australia restricts lithium: lithium price rises, nickel and cobalt prices fall

2. Unilateral policy: Australian lithium and geopolitics



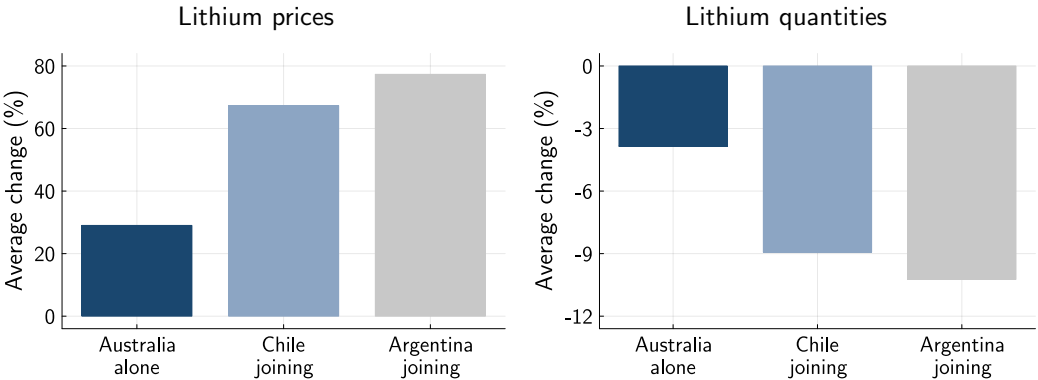
- Australia restricts lithium: substitutes win and free-ride, complements lose

2. Unilateral policy: green adoption and joint use



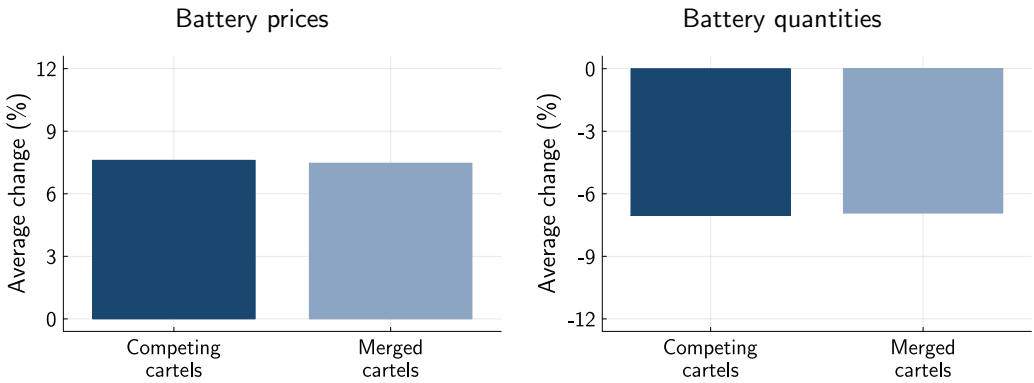
- Lithium is common input: joint use worsens losses (no substitute)
- Nickel is complement: joint use blunts losses (L subsidy)

3. Cartel policy: OPEC for lithium



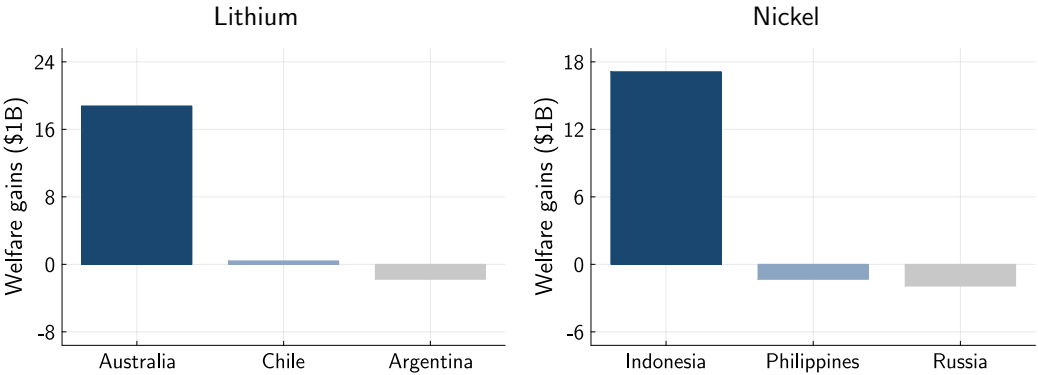
- Cartel restricts lithium: bigger cut, bigger effects

3. Cartel policy: OPEC for nickel and lithium



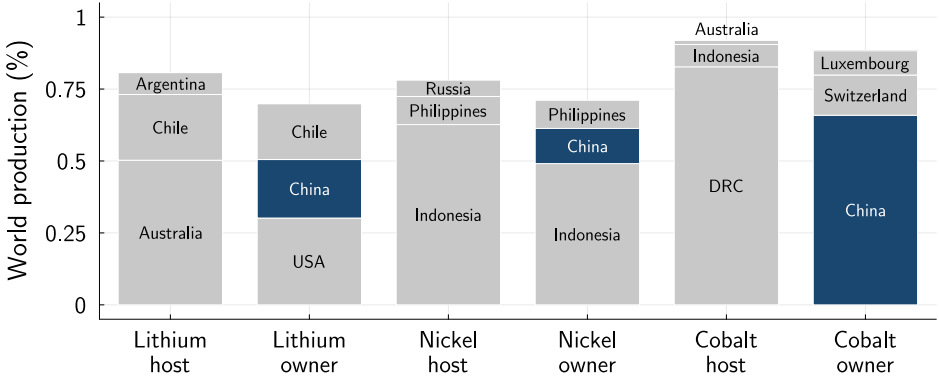
- Consolidation vs. complementarity (Cournot 1838)

3. Cartel policy: cohesion



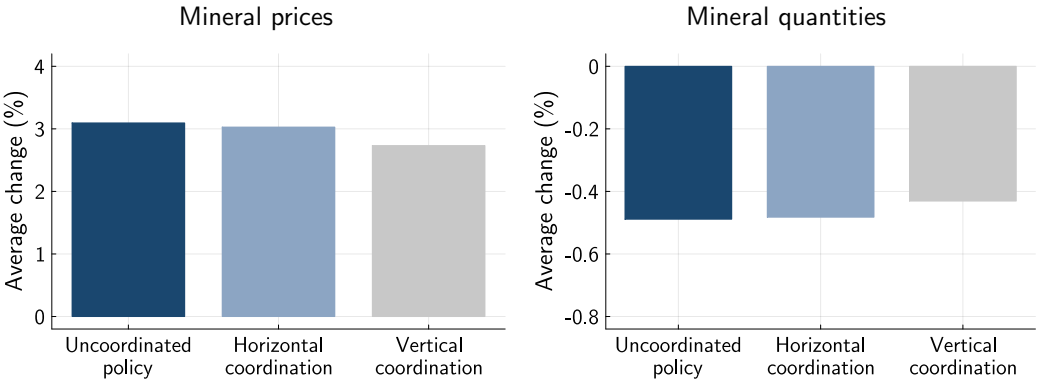
- Australia and Indonesia are anchors, but most others prefer to defect

4. Chinese policy: ownership



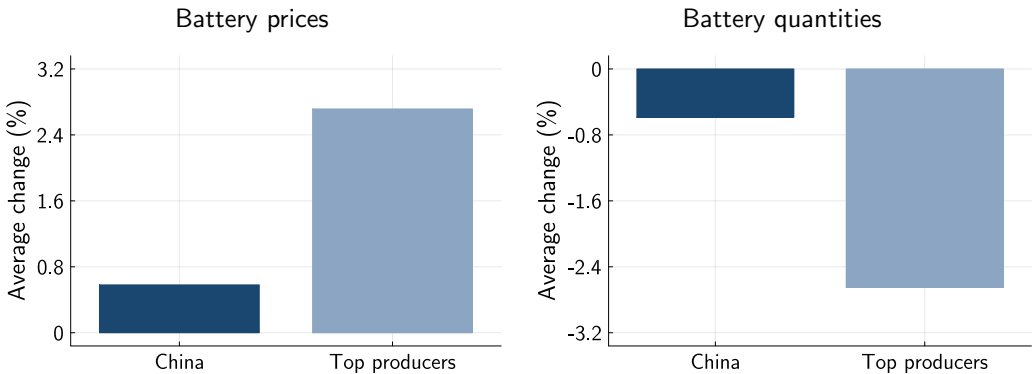
- Chinese companies own mines across minerals

4. Chinese policy: coordination



- Horizontal coordination internalizes cross-mineral complementarity
- Vertical coordination accounts for downstream battery consumption

5. Chinese policy: resource nationalism



- Chinese multinationals vs. independent Australian and Indonesian policy
  - Top producers have larger market shares and do not coordinate across minerals

Summary

- Critical minerals are the next generation of energy resources
- These markets are **concentrated** and **interdependent**
  - With implications for geopolitics and green adoption